

# Rodrigue GOVAN, Ph.D.

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## Education

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- October 2021 – August 2025    **Ph.D. in Computer Science – Data Science**  
Doctoral School in the Pacific (ED469), University of New Caledonia
- Thesis title: “**Deep Learning on Attributed Graphs for Mapping Leptospirosis Risk**”.
  - Supervision:
    - Nazha SELMAOUI-FOLCHER, Full Professor in Computer Science in the Institute of Exact and Applied Sciences at the University of New Caledonia;
    - Philippe FOURNIER-VIGER, Distinguished Professor in Computer Science in the Big Data Institute at Shenzhen University, China.
  - Fundings: Doctoral scholarship (50%), ANR-SPIraL (ANR-19-CE35-0006-02, 50%).
  - Jury members:
    - Christophe MENKÈS [**Chair**], Sr. Research Scientist in Climatology at ENTROPIE, French National Institute of Research for Development (IRD, New Caledonia);
    - Thomas GUYET [**Reviewer**], Full Researcher in Computer Science at AIstroSight, Inria Research Centre, Lyon, France;
    - Luiz-Angelo STEFFENEL [**Reviewer**], Full Professor in Computer Science at LICIS, University of Reims Champagne-Ardenne;
    - Cyrille GOARANT [**Examiner**], Research Scientist in Epidemiology at the Public Health Department, South Pacific Community;
    - Corina IOVAN [**Examiner**], *Research Scientist* in Ecology and Artificial Intelligence at ENTROPIE, French National Institute of Research for Development (IRD);
    - Nadia KABACHI [**Examiner**], Full Professor in Computer Science at ERIC, University Claude Bernard Lyon 1, France.
  - Access to thesis manuscript (*in French*) : [hal.science/tel-05268357](https://hal.science/tel-05268357)
- 2017 – 2019    **M.Sc. in Applied Mathematics and Statistics**, University of Bordeaux
- Minor: Statistical and Stochastic Modeling.
  - Final-year project: “**Object detection in satellites images using deep learning**”.
- 2014 – 2017    **B.Sc. in Applied Mathematics and Computer Science**, University of Bordeaux
- Minor: Business Administration.
  - Final-year project: “**Object-oriented programming of a role-playing game in Java**”.

## Experiences

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- September 2020 – December 2020    **Data Scientist**, Institute of Exact and Applied Sciences, UNC – Noumea, New Caledonia
- Modeling of data related to the quality of freshwater and groundwater in New Caledonia, in collaboration with the water service of the department of veterinary, food and rural affairs of New Caledonia.
  - Descriptive and inferential analyses of physicochemical data.
  - Implementation of a graphical interface (using Python and **dash**) for the visualization of the analyses.

- Deployment of the interface on a cloud computing platform.
- Contributions to the writing and publication of research articles.
- Tools used: Python (**sklearn**, **statsmodels**, **plotly**, **Jupyter**), QGIS, Microsoft Access, PostgreSQL, L<sup>A</sup>T<sub>E</sub>X.

- August 2019 – February 2019    **Data Scientist Intern**, IDAIA Group – Bruges (Bordeaux), France
- Detection of residential swimming pools in satellite images using deep learning to complete the French land registry.
  - Labeling of satellite images, creation of training datasets, and training of object detection models (Faster R-CNN, MobileNet, YOLO).
  - Use of satellite images from the French National Institute of Geographic and Forest Information.
  - Over 200M images (covering the entire country of France) processed.
  - Parallelization of predictions on CPU and GPU.
  - Estimation of the surface area of detected swimming pools using clustering methods.
  - Tools used: Python (**sklearn**, **Tensorflow**, **PyTorch**, **plotly**, **tensorboard**, **WandB.ai**, **Jupyter Notebook**), QGIS, Linux, L<sup>A</sup>T<sub>E</sub>X.

## Teaching activities

### Teachings

| Course name                             | Level                              | Tutorials | Labs | Total | Years            |
|-----------------------------------------|------------------------------------|-----------|------|-------|------------------|
| Algorithms and Programming II in Python | B.Sc. CS 1 <sup>st</sup> year      | ×         | 34   | 34h   | 2022             |
| Graph theory and Algorithms             | B.Sc. CS 2 <sup>nd</sup> year      | 12h       | 48h  | 60h   | 2022, 2023, 2024 |
| Advanced databases                      | B.Sc. CS 2 <sup>nd</sup> year      | ×         | 48h  | 48h   | 2022, 2023, 2024 |
| Introduction to data analysis           | B.Sc. CS 3 <sup>rd</sup> year      | 4h        | ×    | 4h    | 2023             |
| Applied statistics for geosciences      | B.Sc. Geology 1 <sup>st</sup> year | 8h        | ×    | 8h    | 2024             |
| Total : 154h                            |                                    |           |      |       |                  |

### Internship supervision

- November 2024 – January 2025    **Front-End Developer** – Waren DERAMANE (2<sup>nd</sup> year B.Sc. in Computer Science)
- Design and development of a graphical user interface for managing and visualizing spatio-temporal data.
  - Tools used: Python (**Flask**), HTML, CSS, JavaScript, PostgreSQL, Docker.
  - Supervision: 80% R. GOVAN, 20% N. SELMAOUI-FOLCHER.
- November 2024 – January 2025    **Back-End Developer** – Saïden TIDJINE (2<sup>nd</sup> year B.Sc. in Computer Science)
- Design and development of a spatio-temporal database to support data analysis and management.
  - Tools used: Python (**FastAPI**), PostgreSQL/PostGIS, Docker.
  - Supervision: 80% R. GOVAN, 20% N. SELMAOUI-FOLCHER.
- October 2022 – February 2023    **Tutored project, Data Analyst** – Mathilde NYIKEINE, Jean-François QUINQUIS et Duc Huy Dimitri TRAN (2<sup>nd</sup> year M.Sc. Computer Science Applied to Business Management)
- Spatio-temporal analysis of physico-chemical data.
  - Development of a QGIS-integrated tool for displaying the analyzed data.
  - Tools used: QGIS, PostgreSQL/PostGIS, Python.
  - Supervision: 40% R. GOVAN, 60% N. SELMAOUI-FOLCHER.

## Technical skills

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|---------------------------------|--------------------------------------------------------------------------------------------------|
| Operating systems               | macOS, Linux (Ubuntu, Raspberry Pi OS, Debian), Windows                                          |
| Management and deployment tools | SLURM workload manager, Docker, GitHub/GitLab                                                    |
| Databases                       | MySQL, PostgreSQL/PostGIS                                                                        |
| Programming languages           | Python, R, MATLAB/Scilab, Java, HTML, CSS, SQL, PHP, Shell Unix, L <sup>A</sup> T <sub>E</sub> X |
| Specific librairies             | PyTorch, scikit-learn, dash, rasterio                                                            |

## Projects

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- 18–20 June 2021    **Hackathon (CASSINI.eu, 1<sup>st</sup> Edition)** – University of New Caledonia
- European entrepreneurship-oriented Hackathon.
  - Hackathon’s theme: “Digitizing Green Spaces”.
  - Proposed project: **SANDLESS (Software to Analyze Natural Data to Lower Ecological Stress with Satellites)**, a software analyzing satellites, social, economical and environmental data of cities from all over the world in order to help decision-makers to build their own city.
  - French rank: 1<sup>st</sup>.
  - European (final) rank: 3<sup>rd</sup>.
- October 2018    **Big Data Project** – University of Bordeaux  
– January 2019
- Kaggle-like contest organized by the INSA-Toulouse engineering school in collaboration with Airbus Defence and Space.
  - Contest including universities and engineering schools in France.
  - Objective: To determine, by using satellite images, whether or not the image contains a wind turbine.
  - Tools used: Python, Tensorflow, Keras.
  - Final rank: 7/64 teams (with a 97.7% accuracy).

## Publications and Communications

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### Journal papers with peer review

- [3]    **Govan, R.**, Scherrer, R., Fougeron, B., Laporte-Magoni, C., Thibeaux, R., Genthon, P., Fournier-Viger, P., Goarant, C., and Selmaoui-Folcher, N. (2025). **Spatio-temporal risk prediction of leptospirosis: A machine-learning-based approach**. *PLOS Neglected Tropical Diseases*, 19(1), e0012755.  
DOI: [10.1371/journal.pntd.0012755](https://doi.org/10.1371/journal.pntd.0012755)
- [2]    Thibeaux, R., Genthon, P., **Govan, R.**, Selmaoui-Folcher, N., Tramier, C., Kainiu, M., Soupé-Gilbert, M.-E., Wijesuriya, K., and Goarant, C. (2024). **Rainfall-driven resuspension of pathogenic *Leptospira* in a leptospirosis hotspot**. *Science of The Total Environment*, 911, 168700.  
DOI: [10.1016/j.scitotenv.2023.168700](https://doi.org/10.1016/j.scitotenv.2023.168700)
- [1]    Scherrer, R., **Govan, R.**, Quiniou, T., Jauffrais, T., Lemonnier, H., Bonnet, S., and Selmaoui-Folcher, N. (2022). **Real-Time Automatic Plankton Detection, Tracking and Classification on Raw Hologram**. In: *International Meeting on Computational Intelligence Methods for Bioinformatics and Biostatistics* (pp. 25-39). Springer, Cham.  
DOI: [10.1007/978-3-031-20837-9\\_3](https://doi.org/10.1007/978-3-031-20837-9_3)

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## International conferences with peer review

- [5] **Govan, R.**, Scherrer, R., Fournier-Viger, P., and Selmaoui-Folcher, N. (2025). **SPAPOOL: Soft Partition Assignment Pooling for Graph Neural Networks**. In: Leung, C.K., Dignös, A., Kotsis, G., Tjoa, A.M., Khalil, I. (eds) *Big Data Analytics and Knowledge Discovery. DaWaK 2025*. Lecture Notes in Computer Science, vol 16048. Springer, Cham.  
DOI: [10.1007/978-3-032-02215-8\\_27](https://doi.org/10.1007/978-3-032-02215-8_27)
- [4] **Govan, R.**, Selmaoui-Folcher, N., Giannakos, A., and Fournier-Viger, P. (2023). **Co-location Pattern Mining Under the Spatial Structure Constraint**. In: Strauss, C., Amagasa, T., Kotsis, G., Tjoa, A.M., Khalil, I. (eds) *Database and Expert Systems Applications. DEXA 2023*. Lecture Notes in Computer Science, vol 14146. Springer, Cham.  
DOI: [10.1007/978-3-031-39847-6\\_13](https://doi.org/10.1007/978-3-031-39847-6_13)
- [3] Tokotoko, J., **Govan, R.**, Lemonnier, H., and Selmaoui-Folcher, N. (2022). **Multiscale and Multivariate Time Series Clustering: A New Approach**. In: Ceci, M., Flesca, S., Masciari, E., Manco, G., Raś, Z.W. (eds) *Foundations of Intelligent Systems. ISMIS 2022*. Lecture Notes in Computer Science(), vol 13515. Springer, Cham.  
DOI: [10.1007/978-3-031-16564-1\\_27](https://doi.org/10.1007/978-3-031-16564-1_27)
- [2] Scherrer, R., **Govan, R.**, Quiniou, T., Jauffrais, T., Lemonnier, H., Bonnet, S., and Selmaoui-Folcher, N. (2021). **Automatic Plankton Detection and Classification on Raw Hologram with a Single Deep Learning Architecture**. In: *CIBB 2021 Computational Intelligence Methods for Bioinformatics and Bio-statistics*.  
URL: [hal-03565469](https://hal.archives-ouvertes.fr/hal-03565469)
- [1] Tokotoko, J., Selmaoui-Folcher, N., **Govan, R.**, and Lemonnier, H. (2021). **TSX-Means: An Optimal K Search Approach for Time Series Clustering**. In: Strauss, C., Kotsis, G., Tjoa, A.M., Khalil, I. (eds) *Database and Expert Systems Applications. DEXA 2021*. Lecture Notes in Computer Science(), vol 12924. Springer, Cham.  
DOI: [10.1007/978-3-030-86475-0\\_23](https://doi.org/10.1007/978-3-030-86475-0_23)

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## National conferences with peer review

- [3] **Govan, R.**, Scherrer, R., Fournier-Viger, P., and Selmaoui-Folcher, N. (2025). **Pooling de Graph Neural Networks : une approche dense mais adaptative**. In: *CNIA 2025-Conférence Nationale en Intelligence Artificielle*, PFIA (No. 55-63).  
URL: [hal-05197596v1](https://hal.archives-ouvertes.fr/hal-05197596v1)
- [2] **Govan, R.**, Scherrer, R., Goarant, C., Cannet, A., Fournier-Viger, P., and Selmaoui-Folcher, N. (2025). **Cartographie du risque épidémiologique : Le défi des données fortement déséquilibrées**. In: *Revue des Nouvelles Technologies de l'Information, 25èmes Journées Francophones Extraction et Gestion des Connaissances, EGC 2025*, vol. RNTI-E-41. (pp. 159-170).  
URL: [hal-04945686](https://hal.archives-ouvertes.fr/hal-04945686)
- [1] **Govan, R.**, Selmaoui-Folcher, N., Giannakos, A., and Fournier-Viger, P. (2023). **Extraction de co-localisations sous contrainte de la structure spatiale**. In: *CNIA 2023-Conférence Nationale en Intelligence Artificielle*, PFIA (No. 53-61).  
URL: [hal-04945686](https://hal.archives-ouvertes.fr/hal-04945686)

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## Posters and workshops

- [2] Genthon, P., Thibeaux, R., Selmaoui-Folcher, N., **Govan, R.**, Kainiu, M., Soupé-Gilbert, M.-E., and Goarant, C. (2025). **Leptospirosis: a critical zone disease?** In : *3<sup>rd</sup> OZCAR TERENO International Conference, Advancing Critical Zone Science*. (Vol. 2025, pp. S14-P2).  
URL: [hal-05387574](https://hal.archives-ouvertes.fr/hal-05387574)

- [1] Genthon, P., Thibeaux, R., Selmaoui-Folcher, N., **Govan, R.**, Kainiu, M., Soupé-Gilbert, M. E., and Goarant, C. (2024). **Leptospira in Rivers of a Leptospirosis Hotspot: Scale Effects**. In: *AGU Fall Meeting Abstracts* (Vol. 2024, No. 1055, pp. H13D-1055).  
URL: [2024AGUFMH13D.1055G](#)

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## Project reports and thesis

- [2] **Govan, R.** (2025). **Deep Learning on Attributed Graphs for Mapping Leptospirosis Risk**. Ph.D. thesis. University of New Caledonia.  
URL: [tel-05268357](#)
- [1] **Govan, R.** (2019). **Deep Learning: Residential pool detection in France**. M.Sc. thesis. University of Bordeaux, France.

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## Other communications

- [3] **Govan, R.**, Parmentier, J.-B., and Quiniou T. (2025). **Discover an interactive 3D model of New Caledonia**. In : Nuit de la Science, Fête de la Science 2025. Université de la Nouvelle-Calédonie.
- [2] **Govan, R.** (2023). **La recherche scientifique au service de la data**. In : *7<sup>ème</sup> meetup consacré à la data*. ISI-NC, OoTech. Nouvelle-Calédonie.
- [1] **Govan, R.** (2022). **Graphes attribués dynamiques et évolutifs : Application à la Cartographie du Risque de Leptospirose en Nouvelle-Calédonie**. In : *15<sup>ème</sup> Édition des Doctoriales*. École Doctorale du Pacifique (ED469). Université de la Nouvelle-Calédonie.

## Professional and academic references

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|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prof. Nazha SELMAOUI-FOLCHER  | Full Professor in Computer Science. Institute of Exact and Applied Sciences, University of New Caledonia. Ph.D. supervisor.<br>E-mail: <a href="mailto:nazha.selmaoui@unc.nc">nazha.selmaoui@unc.nc</a>                                  |
| Prof. Philippe FOURNIER-VIGER | Distinguished Professor in Computer Science. Big Data Institute, College of Computer Science and Software Engineering, Shenzhen University, China. Ph.D. supervisor.<br>E-mail: <a href="mailto:philfv@szu.edu.cn">philfv@szu.edu.cn</a> |
| Prof. Jérémie BIGOT           | Full Professor in Applied Mathematics. Institute of Mathematics in Bordeaux, University of Bordeaux, France. M.Sc. supervisor.<br>E-mail: <a href="mailto:jeremie.bigot@math.u-bordeaux.fr">jeremie.bigot@math.u-bordeaux.fr</a>         |